

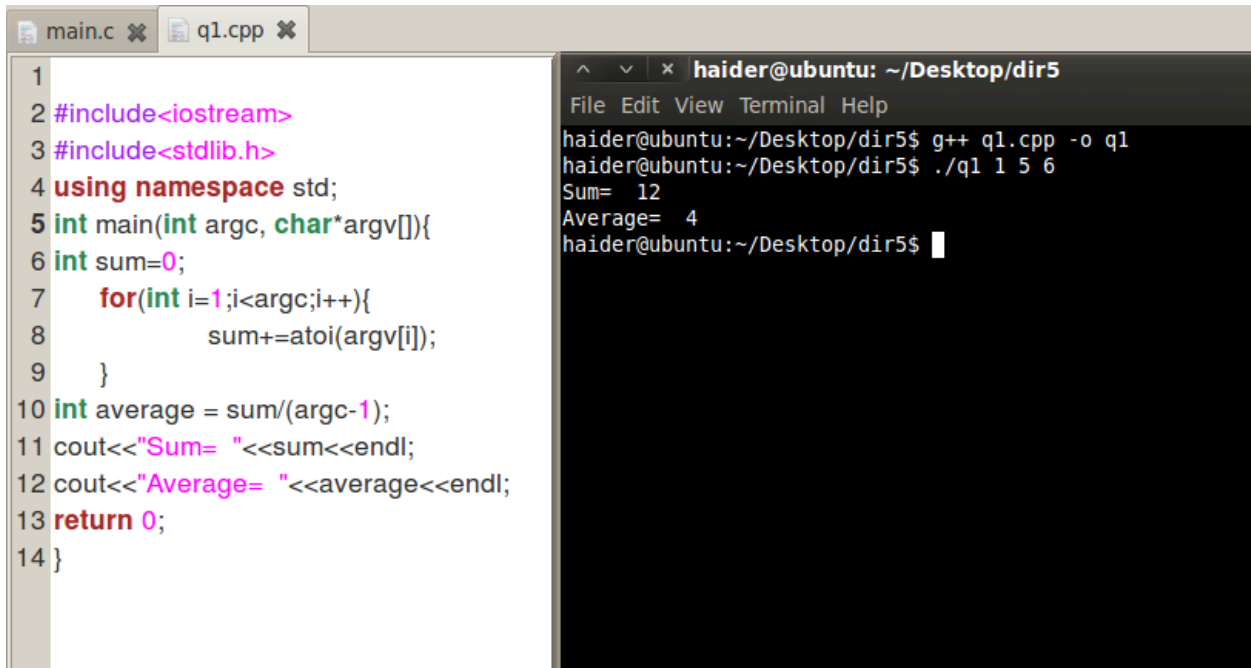
Lab 2

Q1

Argc (argument count), argv(vector)

./q1 1 2 3 (4 in argc)

[0] [1] [2] [3] in argv



```
1
2 #include<iostream>
3 #include<stdlib.h>
4 using namespace std;
5 int main(int argc, char*argv[]){
6     int sum=0;
7     for(int i=1;i<argc;i++){
8         sum+=atoi(argv[i]);
9     }
10    int average = sum/(argc-1);
11    cout<<"Sum= "<<sum<<endl;
12    cout<<"Average= "<<average<<endl;
13    return 0;
14 }
```

```
haider@ubuntu: ~/Desktop/dir5
File Edit View Terminal Help
haider@ubuntu:~/Desktop/dir5$ g++ q1.cpp -o q1
haider@ubuntu:~/Desktop/dir5$ ./q1 1 5 6
Sum= 12
Average= 4
haider@ubuntu:~/Desktop/dir5$
```

Q2

The image shows a C++ IDE with two tabs: q1.cpp and q2.cpp. The q2.cpp tab is active, displaying a bubble sort program. The code is as follows:

```
1 #include<iostream>
2 #include<stdlib.h>
3 using namespace std;
4 int main(int argc, char*argv[]){
5     int size =argc-1;
6     int array[size];
7     for(int i=0;i<size;i++){
8         array[i]=atoi(argv[i+1]);
9     }
10    for(int i=0;i<size;i++){
11        for(int j=i+1; j<size; j++){
12            if(array[i]>array[j]){
13                int temp = array[i];
14                array[i]=array[j];
15                array[j]=temp;
16            }
17        }
18    }
19    cout<<"Sorted Array= ";
20    for(int i=0; i<size;i++){
21        cout << array[i] << " ";
22    }
23    return 0;
24 }
```

To the right of the IDE is a terminal window titled "haider@ubuntu: ~/Desktop/dir5". It shows the compilation and execution of the program:

```
haider@ubuntu:~/Desktop/dir5$ g++ q2.cpp -o q2
q2.cpp: In function 'int main(int, char**)':
q2.cpp:13: error: 'temp' was not declared in this scope
haider@ubuntu:~/Desktop/dir5$ g++ q2.cpp -o q2
haider@ubuntu:~/Desktop/dir5$ ./q2 5 6 9 3 8 4
Sorted Array= 3 4 5 6 8 9 haider@ubuntu:~/Desktop/dir5$
```

Q4

The image shows a C++ IDE with two tabs: file.txt and q4.cpp. The q4.cpp tab is active, displaying a program that reads the contents of a file. The code is as follows:

```
1 #include<iostream>
2 #include<fstream>
3 #include<string>
4 #include<stdlib.h>
5 using namespace std;
6 int main(int argc, char* argv[]){
7
8     ifstream mFile(argv[1]);
9     string data;
10    while(getline(mFile,data)){
11        cout<<endl;
12        cout<<"Contents in txt File: "<<data<<endl;
13    }
14    mFile.close();
15    return 0;
16 }
```

To the right of the IDE is a terminal window titled "haider@ubuntu: ~/Desktop/dir5". It shows the compilation and execution of the program:

```
haider@ubuntu:~/Desktop/dir5$ g++ q4.cpp -o q4
haider@ubuntu:~/Desktop/dir5$ ./q4 file.txt

Contents in txt File: Operating Systems Lab

Contents in txt File: Mudassar Hussian
haider@ubuntu:~/Desktop/dir5$
```

Q3

-o executable file

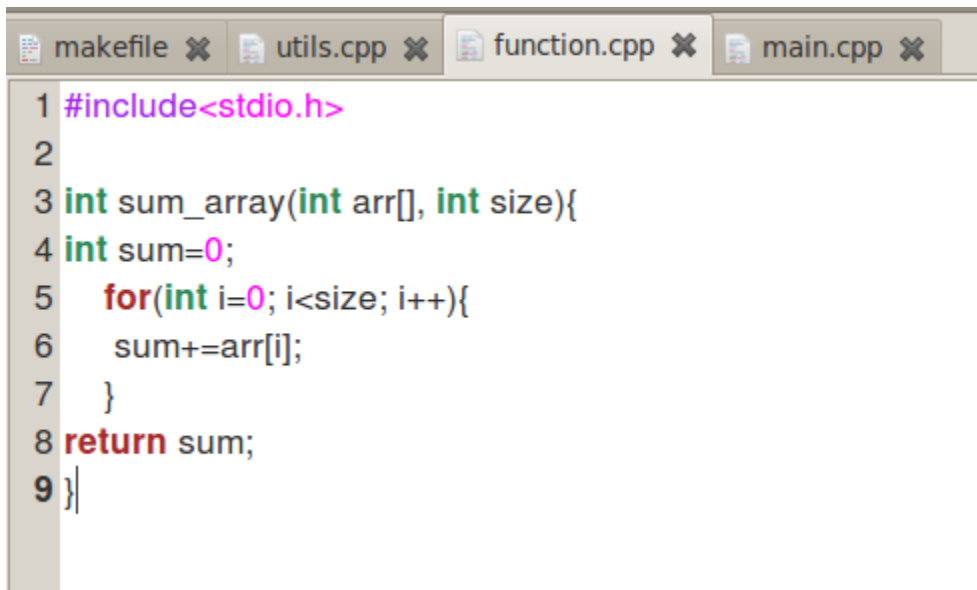
-c object file

What is make file?

gcc -c eval.cpp

gcc -c main.cpp

function.cpp



```
1 #include<stdio.h>
2
3 int sum_array(int arr[], int size){
4     int sum=0;
5     for(int i=0; i<size; i++){
6         sum+=arr[i];
7     }
8     return sum;
9 }
```

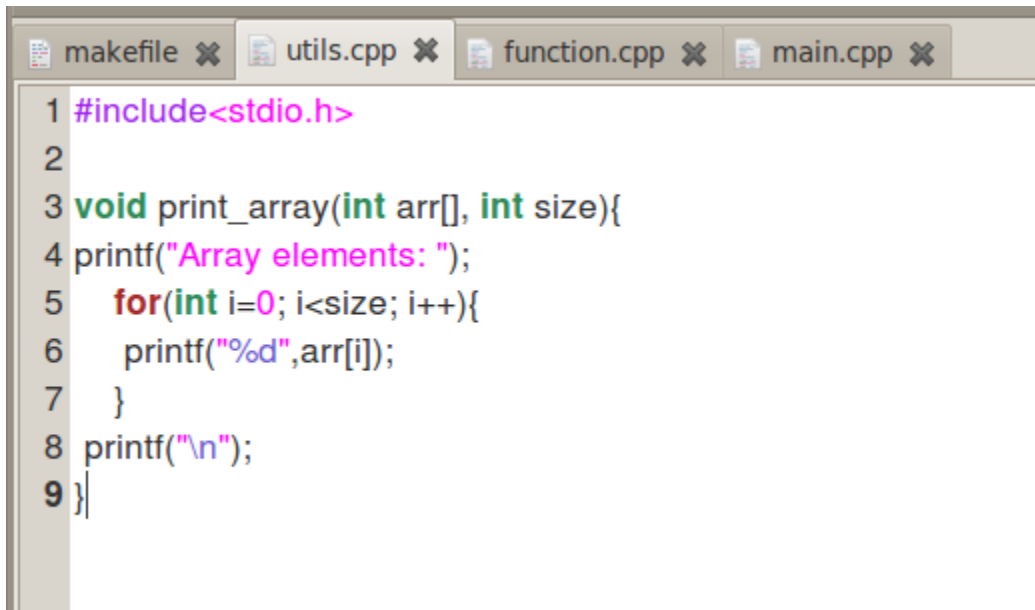
main.cpp

```
makefile x utils.cpp x function.cpp x main.cpp x
1 #include <stdio.h>
2 int sum_array(int arr[], int size);
3 void print_array(int arr[], int size);
4 int main(){
5
6     int arr[]={1,2,3,4,5};
7     int size=5;
8     print_array(arr,size);
9     int sum = sum_array(arr,size);
10    printf("Sum of array elements= %d\n", sum);
11    return 0;
12 } |
```

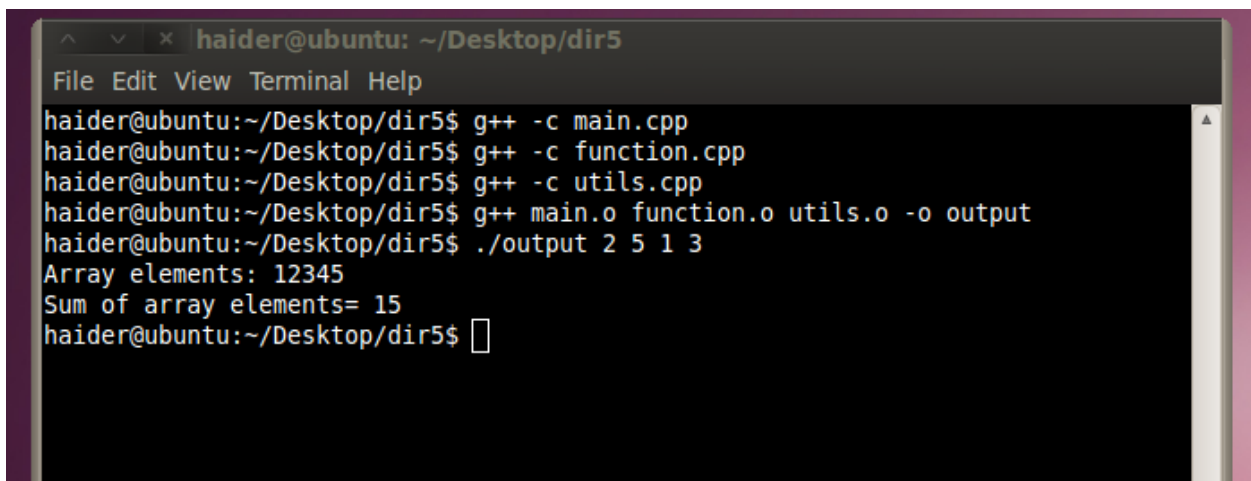
Makefile

```
makefile x utils.cpp x function.cpp x main.cpp x
1 all: output
2 output: main.o function.o utils.o
3 g++ main.o function.o utils.o -o output
4
5 main.o: main.c
6 g++ -c main.c
7
8 function.o: function.c
9 g++ -c function.c
10
11 utils.o: utils.c
12 g++ -c utils.c
13 |
```

utils.cpp



```
1 #include<stdio.h>
2
3 void print_array(int arr[], int size){
4     printf("Array elements: ");
5     for(int i=0; i<size; i++){
6         printf("%d",arr[i]);
7     }
8     printf("\n");
9 }
```



```
haider@ubuntu: ~/Desktop/dir5
File Edit View Terminal Help
haider@ubuntu:~/Desktop/dir5$ g++ -c main.cpp
haider@ubuntu:~/Desktop/dir5$ g++ -c function.cpp
haider@ubuntu:~/Desktop/dir5$ g++ -c utils.cpp
haider@ubuntu:~/Desktop/dir5$ g++ main.o function.o utils.o -o output
haider@ubuntu:~/Desktop/dir5$ ./output 2 5 1 3
Array elements: 12345
Sum of array elements= 15
haider@ubuntu:~/Desktop/dir5$
```

```
haider@ubuntu: ~/Desktop/dir5
File Edit View Terminal Help
haider@ubuntu:~/Desktop/dir5$ g++ -c main.cpp
haider@ubuntu:~/Desktop/dir5$ g++ -c function.cpp
haider@ubuntu:~/Desktop/dir5$ g++ -c utils.cpp
haider@ubuntu:~/Desktop/dir5$ g++ main.o function.o utils.o -o output
haider@ubuntu:~/Desktop/dir5$ ./output 2 5 1 3
Array elements: 12345
Sum of array elements= 15
haider@ubuntu:~/Desktop/dir5$ g++ main.o function.o utils.o -o output
haider@ubuntu:~/Desktop/dir5$ ./output 5 6 2 3
Array elements: 12345
Sum of array elements= 15
haider@ubuntu:~/Desktop/dir5$ ./output
Array elements: 12345
Sum of array elements= 15
haider@ubuntu:~/Desktop/dir5$
```

Q5

```
file.txt x q4.cpp x q5.cpp x
1 #include<iostream>
2 #include <stdlib.h>
3 using namespace std;
4 int main(int argc, char*argv[]){
5     int size=argc-1;
6     int array[size];
7     for (int i=0; i<size; i++){
8         array[i]= atoi(argv[i+1]); }
9     int a= 0;
10    int b= 1;
11    int c;
12    for(int i=2; i<size;i++){
13        c=a+b;
14        if(array[i]!=c){
15
16            cout<< "missing element is: " << c <<endl;
17            break;
18        }
19        a=b;
20        b=c; }
21    return 0;
22 }
```

```
haider@ubuntu: ~/Desktop/dir5
File Edit View Terminal Help
haider@ubuntu:~/Desktop/dir5$ g++ q5.cpp -o q5
haider@ubuntu:~/Desktop/dir5$ ./q5 0 1 1 3
missing element is: 2
haider@ubuntu:~/Desktop/dir5$
```